

ADAPTIVE COMPETITIVE CODING PLATFORM

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Abstract

Enables personalized coding practice, real-time competition, and data-driven performance tracking in a unified platform

Problem Statement & Objectives

Existing coding platforms lack real-time competition, adaptive difficulty, and classroom analytics in one place.

- * Build competitive coding and real-time 1v1 battles.
- * Implement performance-based adaptive problem recommendations.
- * Provide ratings, XP, leaderboards, and activity analytics.

Existing vs Proposed System

Feature	Existing System	Proposed System
[Feature 1]	Individual solving only	Adds real-time 1v1 battles
[Feature 2]	Manual or basic filters	Adaptive recommendations
[Feature 3]	Limited classroom monitoring	Teacher classrooms & analytics
[Feature 4]	Single, basic rating	Multi-tier ratings & XP progression

Feasibility Study

Technical Feasibility: Built using accessible open-source tech like React, Node.js, and Socket.IO.

Economic Feasibility: Low development cost using open-source tools and free-tier services without special hardware.

Operational Feasibility: Intuitive, role-based platform for students and teachers with modular, easy-to-maintain architecture.

Schedule Feasibility: Feasible to build incrementally, from core features to deployment, within a semester.

Tools & Technologies

Front-End: React.js, HTML5, CSS3, JavaScript

Back-End: Node.js, Express.js, REST APIs, Socket.IO

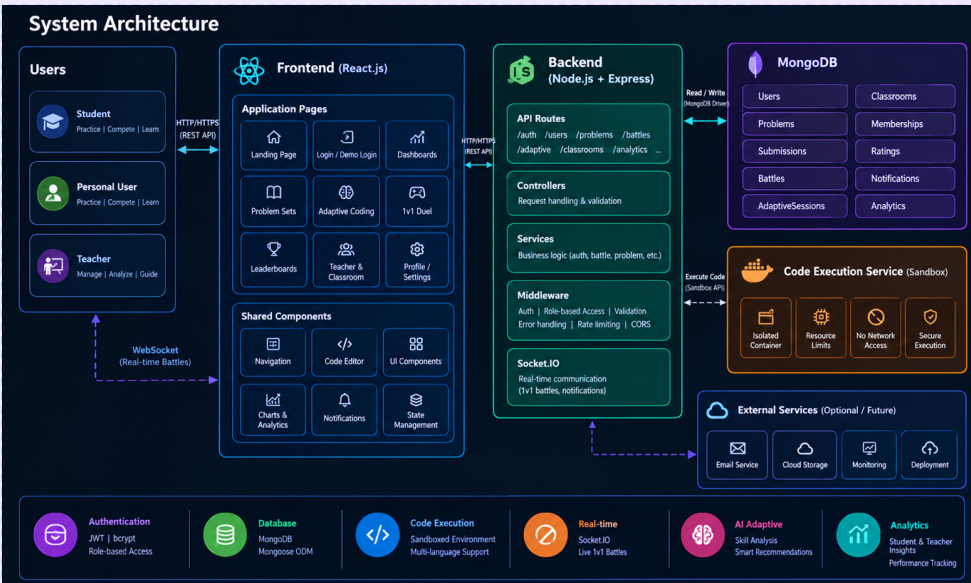
Database: MongoDB with Mongoose

Tools: Git & GitHub, Docker, VS Code, Draw.io

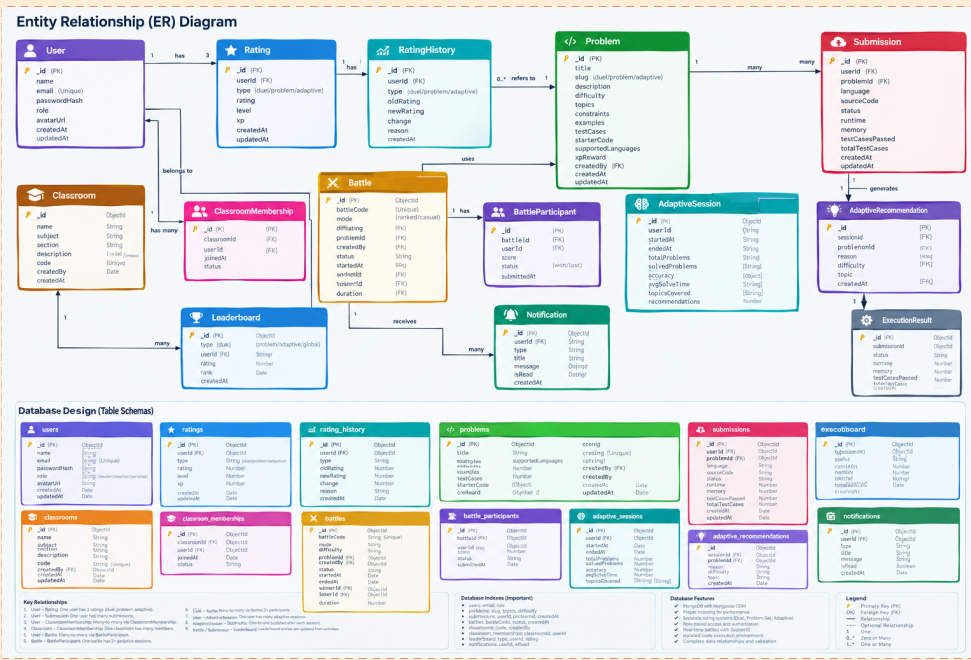
Authentication: JWT, bcrypt

Code Execution: Isolated/Sandboxed Code Execution Environment

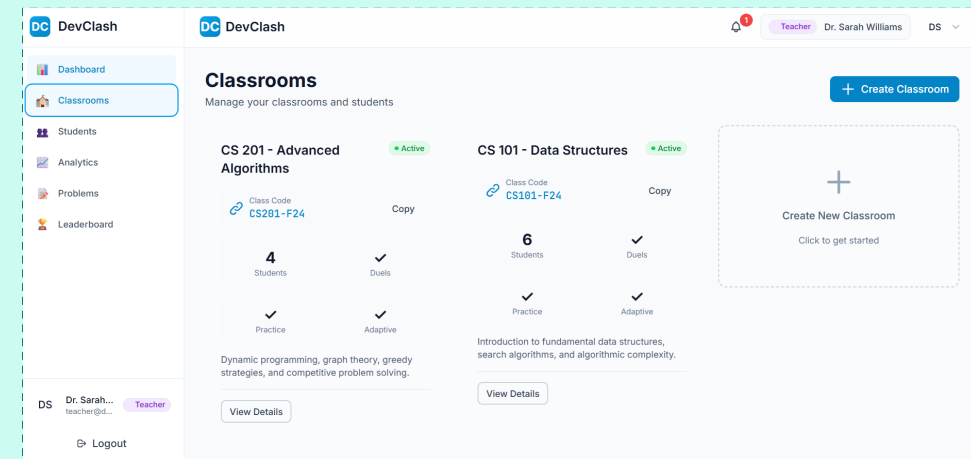
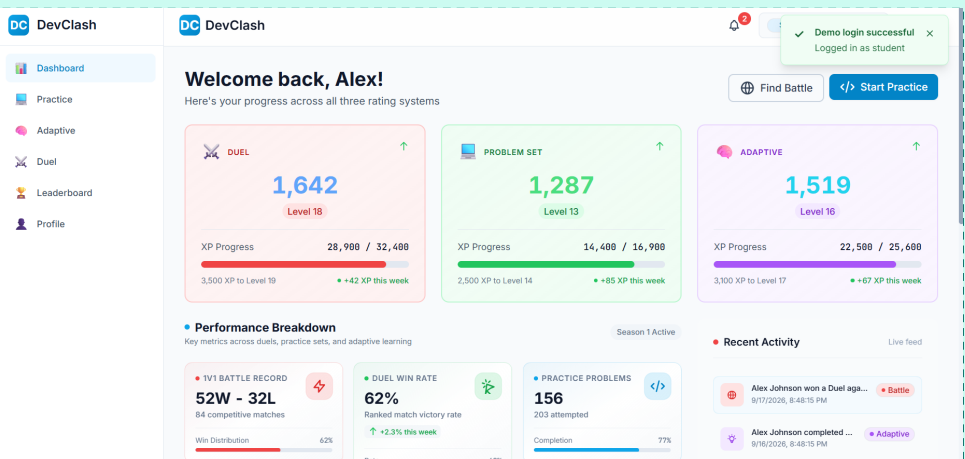
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Results & Testing

Key Outcomes: Integrated coding, 1v1 battles, adaptive practice, ratings, and analytics.

CRUD & Testing: Verified database ops, validation, auth, APIs, and real-time Socket.IO.

Conclusion

Achieved: Built an intelligent coding platform with problem-solving, adaptive learning, 1v1 battles, and analytics.

Objective Fit: Enables 1v1 competition, personalizes practice via adaptation, and tracks multi-activity ratings.

Future Scope

AI Assistant: Add AI code analysis, debugging help, hints, and learning recommendations.

Analytics & Scale: Add skill prediction, plagiarism detection, mobile support, and cloud execution.